Google Cloud H2S HACK2SKILL

AI Builder Cup

Build in JAPAC.
Build for the world.



TEAM DETAILS

TEAM NAME	Mindflayer
TEAM LEADER	Ashfaque Rifaye
TEAM MEMBERS	Ashfaque Rifaye · Vasanth Kumar

PROBLEM STATEMENT

Retail & Commerce - intelligent customer and business experiences: conversational shopping that turns *"I need something that fits"* into the right product or a custom order, from real stores and real makers.

Tesselle
ONE AGENT FOR EVERYTHING THAT HAS TO FIT

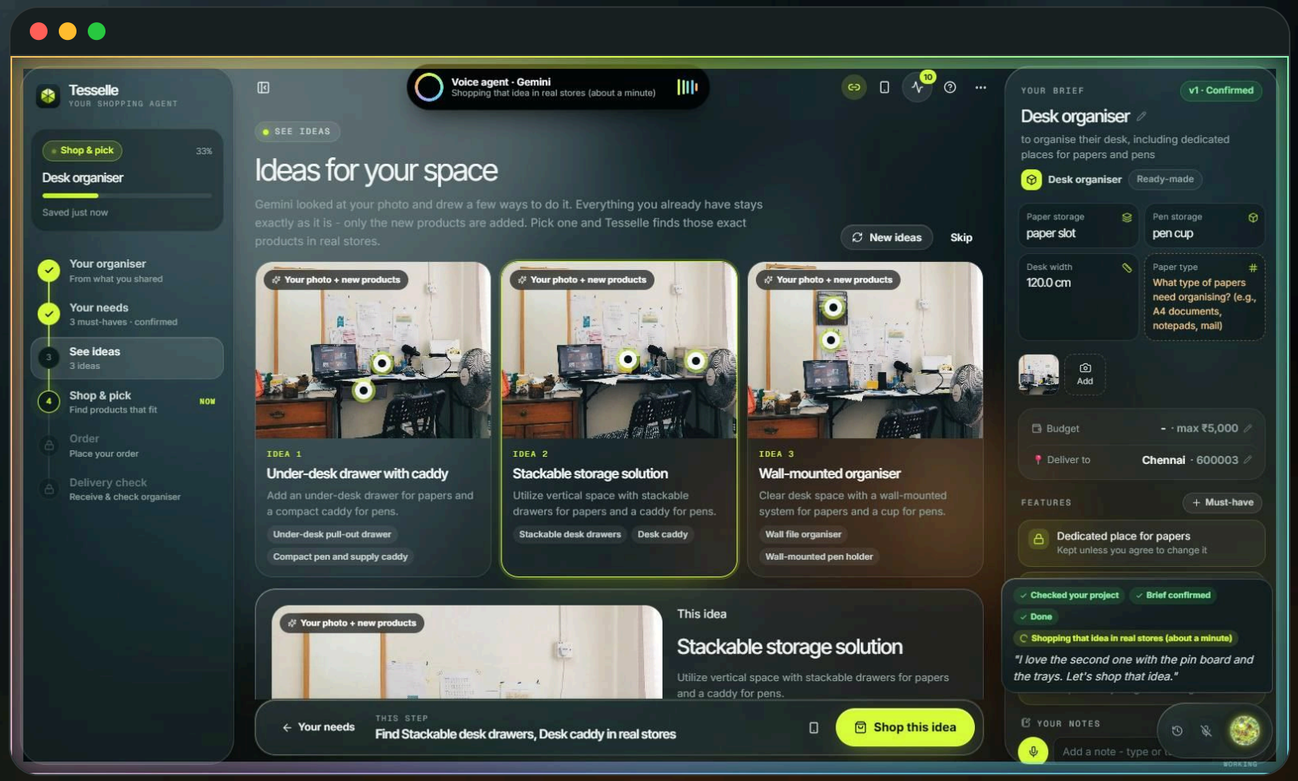
BRIEF ABOUT THE IDEA

One agent for everything that has to fit.

Tesselle is a voice-first shopping agent for things where fit matters - a desk organiser for a cramped desk, a blouse by Friday, 500 printed bags for an event.

You **say it, show it, or look at it** through smart glasses. Gemini writes the brief and the facts that decide fit, **draws ideas into your own photo**, finds real products in stores that deliver to your PIN, **phones and emails real makers**, and fills the store cart with cash on delivery - you place the order.

Every step is checked against your must-haves, and laptop, phone and glasses stay in sync on one project.



OPPORTUNITIES · THE PROBLEM IN NUMBERS

The cost of guessing - and the shift to agents.

$890B

of goods returned to US retailers in 2024

NRF & HAPPY RETURNS · DEC 2024

~70%

of online clothing returns cite size and fit

CORESIGHT RESEARCH · 2026

25–40%

return rate in online fashion; size & fit is the #1 driver

ZALANDO RESEARCH · KDD 2021

$170–180B

India e-retail by 2030, from $65–66B in 2025; 290–300M shoppers

BAIN & FLIPKART · HOW INDIA SHOPS ONLINE 2026

7.2 crore

registered MSMEs - the makers custom orders depend on

MINISTRY OF MSME · LOK SABHA, DEC 2025

$3–5T

global commerce AI agents could orchestrate by 2030; $1T in US retail

MCKINSEY · 2025

Who benefits: shoppers (fewer wrong buys and returns), India's makers and small sellers (qualified, structured orders), and stores (higher conversion, fewer returns).

OPPORTUNITIES

Not a chatbot that suggests links - an agent that does the job.

HOW IS IT DIFFERENT?

- Shopping assistants recommend; Tesselle **acts** - calls makers, finds contacts, fills carts.
- It covers **ready-made and made-to-order** in one journey.
- Ideas are drawn into **your own photo**, not a stock render.

HOW DOES IT SOLVE IT?

- Writes down **the facts that decide fit** and checks every option against them in code.
- Reaches makers **where they work** - phone, WhatsApp, email - and turns answers into quotes.
- Works across glasses, laptop and phone, in sync.

USP

- **Fit as a contract:** must-haves never change without your yes.
- **A real-time AI phone agent** (Gemini Live) that gets quotes from offline makers.
- **Steps planned per purchase** - a desk and 500 bags get different journeys.

Today Search → twenty tabs → guess the size → message makers one by one → return what doesn't fit.

Tesselle One sentence → checked options from real stores and makers → you approve the order or the quote.

LIST OF FEATURES

Everything from "I need" to "it arrived".

Real-time voice agent
Gemini Live drives the whole app; waits for you to finish; resumes where you left off

MetaGlass mode
Simulated smart glasses: Gemini sees what you see, saves the view to the project

Adaptive brief
Must-haves, fit facts and steps planned for each purchase; versioned, restorable

Ideas in your photo
2–3 ideas drawn into your own photo, tagged with the products they need; redraw by voice

Real-store search
Vertex AI Search over 45 Indian stores + Google Search grounding, PIN-aware

Fit checker
Deterministic verdicts for every listing and quote; sellers can propose, only you approve

Shop the look
Real products placed in your photo, each dot linked to its listing

Checkout agent
Fills the store cart, sets the PIN, picks cash on delivery; you place it

Contact finder
Reads each business's own site for its published email and phone

AI phone calls
Gemini Live on a real phone line asks makers for size, price and lead time

Email & WhatsApp
Requests with privacy-scrubbed photos; replies read into checked quotes

Devices in sync
Laptop, phone and glasses mirror step, scroll and agent activity live

PROCESS FLOW

From a sentence to a checked delivery.

YOU
Say · show · look
voice, photo, video, MetaGlass

→

BRIEF WRITER
Brief + fit facts
steps planned for this purchase

→

DESIGN AGENT
Ideas in your photo
pick one, or redraw by voice

→

SHOPPING AGENT + MAKER SCOUT
Real stores & makers
every option fit-checked

READY-MADE PATH

Shop the look
add picks

→

Checkout agent
cart · PIN · COD

→

You place it
on the store

MADE-TO-ORDER / BULK PATH

Contacts found
request written

→

AI call · email
Gemini Live

→

Quotes checked
you approve

Then Order tracked → a photo of what arrived is checked against the brief → accept or report. Any step can be driven by voice or one tap, on any device.

WIREFRAMES / MOCK DIAGRAMS

One project, three screens.



Laptop

Step rail planned per purchase · the step in the centre · the brief on the right · one pinned next action · voice orb.

Phone

Same project as a step strip with a pinned action bar; follows the laptop's step and scroll live.

MetaGlass

Lens view with a heads-up card: what you said, what the agent did, the best fit so far.

ARCHITECTURE DIAGRAM

Built end-to-end on Google Cloud.

CLIENTS
Laptop · Phone · MetaGlass
React web app

FIREBASE HOSTING
CDN · forwards /api

MAKERS & STORES
Twilio phone + WhatsApp
Gmail · store pages

HTTPS · WEBSOCKETS VOICE · CAMERA · CALLS

CLOUD RUN · FASTAPI + GOOGLE ADK
Voice assistant · brief writer & planner · design agent · shopping agent · maker scout · outreach & call agents · checkout agent
Fit checker - deterministic code
Gemini Live bridge for mic and phone audio

FIRESTORE
briefs, options, calls, sync

CLOUD STORAGE
photos, ideas, recordings

SECRET MANAGER
keys & credentials

VERTEX AI ONE SERVICE IDENTITY

GEMINI ON VERTEX AI
Gemini 3 Flash · Gemini Live native audio · Gemini image · Gemini TTS

SEARCH
Vertex AI Search (45 stores) · Grounding with Google Search

AGENT ENGINE
Sessions · Memory Bank

TYPED AGENTS
Every agent returns schema-checked output; the fit checker is plain code, so a model can't wave a misfit through.

SAFE BY DESIGN
Secrets in Secret Manager; photos scrubbed before they leave; calls and email only from the owner's devices.

SHIP & OBSERVE
Cloud Build → Cloud Run; each Gemini call labelled in Cloud Logging, with a daily cost cap.

TECHNOLOGIES USED

Google AI at the core, open tools around it.

GOOGLE AI

- Gemini 3 Flash on Vertex AI (reasoning, vision, video)
- Gemini Live native audio (voice + phone calls)
- Gemini image (ideas, shop the look) · Gemini TTS
- Vertex AI Search · Grounding with Google Search
- Agent Development Kit · Agent Engine Sessions & Memory Bank

GOOGLE CLOUD

- Cloud Run (API, WebSockets)
- Firebase Hosting
- Firestore · Cloud Storage
- Secret Manager · Cloud Build · Artifact Registry
- Cloud Logging (every Gemini call labelled)

APP & INTEGRATIONS

- Python · FastAPI · Pydantic typed outputs
- React · TypeScript · Vite · WebGL orb
- Twilio Media Streams (phone) · WhatsApp
- Gmail SMTP/IMAP
- Demo film: Lyria music, Gemini TTS voices

ESTIMATED IMPLEMENTATION COST

About ₹25–45 per complete journey, near zero when idle.

PER SHOPPER JOURNEY (ESTIMATE)	USD
Gemini text, vision and planning calls	0.02–0.05
Gemini image: 3 ideas + 1 shoppable picture	0.15–0.20
Gemini Live voice, ~3 minutes	0.05–0.10
Vertex AI Search, ~5 queries	~0.02
One AI phone call to a maker, ~2 minutes	0.10–0.15
Total	**≈ 0.30–0.50**

Fixed running cost

Cloud Run scales to zero, Firebase Hosting is free tier, Firestore and Storage stay in single-digit dollars a month at prototype scale.

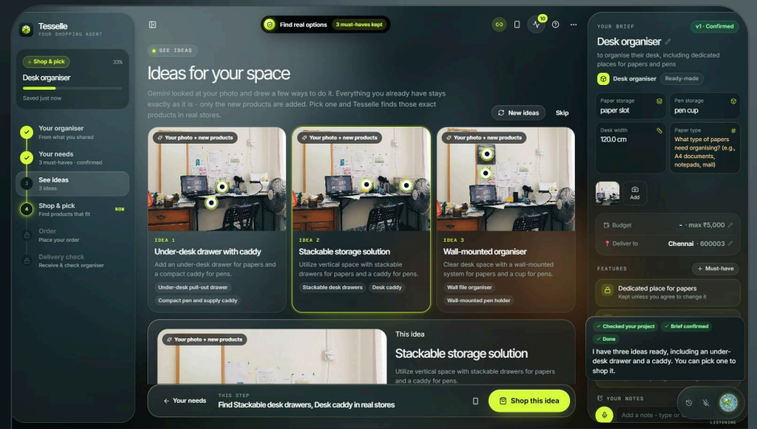
Built-in cost guard

Every Gemini call is labelled; a daily cap limits model and image calls per visitor on the public demo.

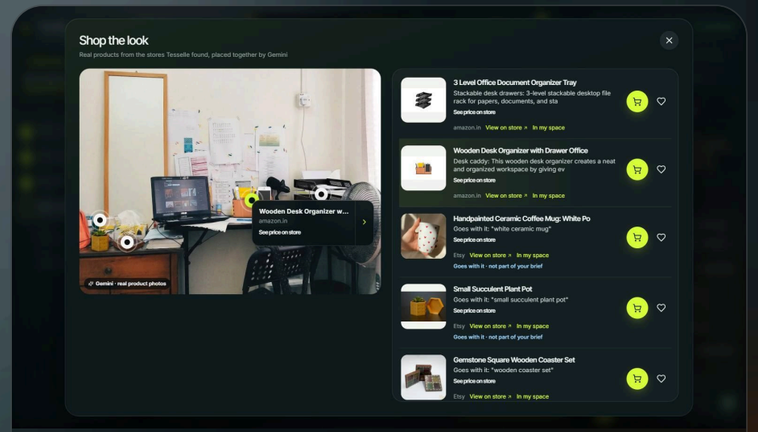
One avoided return or one won custom order pays for many journeys.

Estimates at public list prices, October 2026; actual cost depends on images per journey and call minutes.

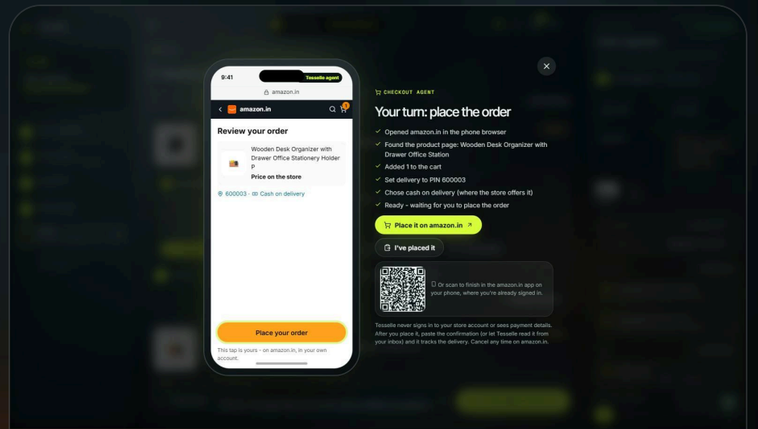
SNAPSHOTS OF THE PROTOTYPE



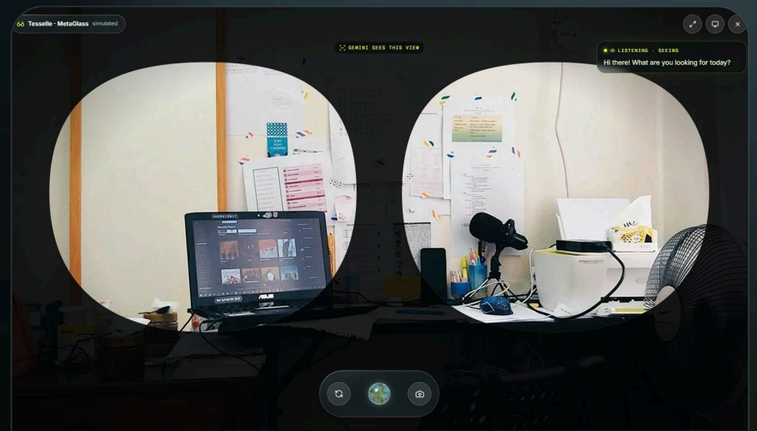
Ideas drawn into your own photo



Shop the look - real, tagged products



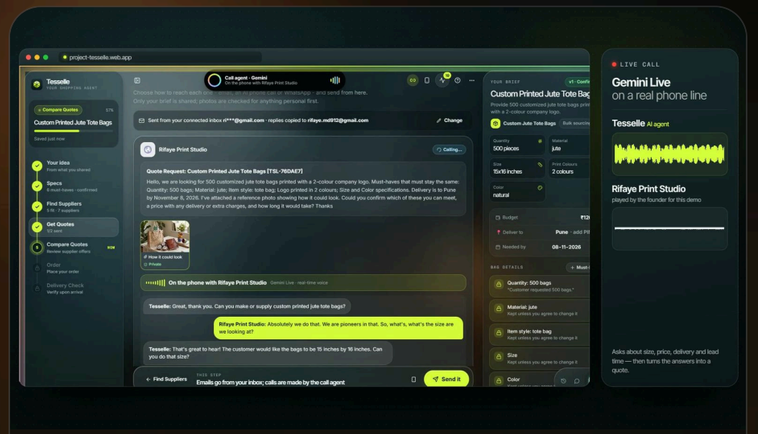
Checkout agent: cart, PIN, cash on delivery



MetaGlass: Gemini Live sees the desk



Steps re-planned for 500 tote bags



A live AI phone call with a print studio

PROTOTYPE PERFORMANCE

Measured on the live app, end to end.

~2.5 s
from you stopping speaking to the voice agent answering

~60 s
to draw 3 ideas into your own photo, each tagged

43
real listings for 3 idea products, plus a shoppable picture, in ~80 s

6
real makers found in parallel with the store search

2.5 min
live AI phone call; 8 answers captured and turned into a checked quote

< 20 s
to find a business's published email and phone on its own site

1 fps
camera frames to Gemini Live in MetaGlass mode

3
devices in sync on one project - step, scroll and agent activity

Single runs on Cloud Run (asia-southeast1), 18 Oct 2026 - indicative, not a load test. Automated end-to-end checks cover brief, search, fit checks, looks, outreach and orders.

ADDITIONAL DETAILS / FUTURE DEVELOPMENT

From one desk to how India buys anything custom.

NOW

Live agent

Ideas in your photo, real stores and makers, live AI calls, checkout agent, three devices in sync.

NEXT

Makers at scale

WhatsApp Business, Indian phone numbers, Hindi and Tamil voices.

THEN

One-tap orders

Agent-ready checkout with store and UPI partners; a real smart-glasses app.

LATER

B2B sourcing

Bulk briefs, samples, pooled demand when minimum quantities block a maker.

Scalability

Stateless Cloud Run instances, Firestore state, per-project WebSockets; agents run in parallel and stream progress.

Business model

Free for shoppers; a disclosed fee on orders and quotes won; subscriptions for makers who want qualified briefs.

LINKS

Try it - no sign-up.

GITHUB PUBLIC REPOSITORY

github.com/ashfaque-rifaye/project-tesselle

DEMO VIDEO LINK (3 MINUTES)

project-tesselle.web.app/demo/tesselle-demo.mp4

FINAL PRODUCT LINK

project-tesselle.web.app

Cloud Run: tesselle-448771854429.asia-southeast1.run.app

USER GUIDE

1. **Start talking** on the homepage (or type) - e.g. "a desk organiser for my messy desk, under ₹5,000".
2. Add a photo of the space when asked; confirm the brief.
3. Say "show me ideas", then "shop the second one".
4. Open **Shop the look**, add picks, and let the checkout agent fill the cart.
5. Or try **MetaGlass mode**, or "Watch it run" for a guided demo.

Scan to open on your phone. Calls and email go out only from the owner's devices.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

